

Estimations Combined

2/28/2024

Description

This code loads and combines the different feature value estimations (natural language writing, paleolithic signs, and proto-cuneiform) into one csv file.

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
#library(stringr)  
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##     filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##     intersect, setdiff, setequal, union
```

Load Data

Load csv files with quantitative feature values for sign sequences.

SIGNBASE

```
# load signbase estimations for strings on objects  
sign.data <- read.csv("~/Github/UpperPalCombinatorics/results/signBase_features.csv")  
# select Aurignacian objects  
sign.data <- sign.data[sign.data$archaeological_unit == "Aurignacien", ]  
  
# select columns for direct comparison with TeDDi sample  
sign.data.selected <- select(sign.data, c('archaeological_unit', 'coding_clean',  
                                           'num.char', 'huni.chars', 'hrate.chars',  
                                           'ttr.chars', 'rm.chars'))  
  
# change column names  
colnames(sign.data.selected) <- c('subcorpus', 'sequence', 'num.char', 'huni.chars', 'hrate.chars',  
                                  'ttr.chars', 'rm.chars')
```

CDLI

```
# read feature values for proto-cuneiform from CDLI
# Uruk V
cuneiV.data <- read.csv("~/Github/UpperPalCombinatorics/results/cuneiform_UrukV_features.csv")

# add column with subcorpus information
cuneiV.data$subcorpus <- rep("Cuneiform (Uruk V)", nrow(cuneiV.data))

# select columns for direct comparison with other samples
cuneiV.data.selected <- select(cuneiV.data, c('subcorpus', 'signs.expand.vec',
                                             'num.char', 'huni.chars', 'hrate.chars',
                                             'ttr.chars', 'rm.chars'))

# rename column with sign sequences
names(cuneiV.data.selected)[names(cuneiV.data.selected) == "signs.expand.vec"] <- "sequence"

# Uruk IV
cuneiIV.data <- read.csv("~/Github/UpperPalCombinatorics/results/cuneiform_UrukIV_features.csv")

# add column with subcorpus information
cuneiIV.data$subcorpus <- rep("Cuneiform (Uruk IV)", nrow(cuneiIV.data))

# select columns for direct comparison with other sample
cuneiIV.data.selected <- select(cuneiIV.data, c('subcorpus', 'signs.expand.vec',
                                             'num.char', 'huni.chars', 'hrate.chars',
                                             'ttr.chars', 'rm.chars'))

# rename column with sign sequences
names(cuneiIV.data.selected)[names(cuneiIV.data.selected) == "signs.expand.vec"] <- "sequence"

# Uruk III
cuneiIII.data <- read.csv("~/Github/UpperPalCombinatorics/results/cuneiform_UrukIII_features.csv")

# add column with subcorpus information
cuneiIII.data$subcorpus <- rep("Cuneiform (Uruk III)", nrow(cuneiIII.data))

# select columns for direct comparison with other sample
cuneiIII.data.selected <- select(cuneiIII.data, c('subcorpus', 'signs.expand.vec',
                                             'num.char', 'huni.chars', 'hrate.chars',
                                             'ttr.chars', 'rm.chars'))

# rename column with sign sequences
names(cuneiIII.data.selected)[names(cuneiIII.data.selected) == "signs.expand.vec"] <- "sequence"
```

TeDDI

```
# read feature values for character sequences of the TeDDi sample
teddi.data <- read.csv("~/Github/UpperPalCombinatorics/results/teddi_features.csv")
# add column with subcorpus information
teddi.data$subcorpus <- rep("TeDDi", nrow(teddi.data))
# select columns
teddi.selected <- select(teddi.data, c('subcorpus', 'text', 'num.char', 'huni.chars', 'hrate.chars',
                                       'ttr.chars', 'rm.chars'))
# rename column with sign sequences
```

```
names(teddi.selected)[names(teddi.selected) == "text"] <- "sequence"
```

Combine

Combine the different data frames.

```
estimations.df <- rbind(sign.data.selected, cuneiV.data.selected,
                        cuneiIV.data.selected, cuneiIII.data.selected,
                        teddi.selected)
head(estimations.df)
```

```
##      subcorpus      sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| ///// _ V      12  1.8962406   1.5295563  0.41666667
## 2 Aurignacien      vvvvv      5  0.0000000   0.5711903  0.20000000
## 3 Aurignacien      vvvv vvvv      9  0.5032583   0.9060047  0.22222222
## 4 Aurignacien vvvvvvvvvvvvvv     14  0.0000000   0.6379550  0.07142857
## 5 Aurignacien      |||      3  0.0000000   0.4308271  0.33333333
## 6 Aurignacien      |||      4  0.0000000   0.5070802  0.25000000
##      rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Simple Stats

```
nrow(estimations.df) # number of sequences with feature estimations
```

```
## [1] 2991
```

```
unique(estimations.df$subcorpus) # names of subcorpora
```

```
## [1] "Aurignacien"      "Cuneiform (Uruk V)"  "Cuneiform (Uruk IV)"
## [4] "Cuneiform (Uruk III)" "TeDDi"
```

```
nrow(estimations.df[estimations.df$subcorpus=="Aurignacien",]) # number of Aurignacien sequences
```

```
## [1] 243
```

```
nrow(estimations.df[estimations.df$subcorpus=="TeDDi",]) # number of TeDDi (modern writing) sequences
```

```
## [1] 821
```

```
nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk V)",])
```

```
## [1] 123
```

```
nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk IV)",])
```

```
## [1] 904
```

```
nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk III)",])
```

```
## [1] 900
```

Write to file

```
write.csv(estimations.df, "~/Github/UpperPalCombinatorics/results/features_combined.csv", row.names = F
```